# Data-Backed Journal-Style Replication of *Position and Differentiation of Firms in Technology Space*

Replication Paper Project

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**Abstract**

This paper presents a local calculated replication of Position and Differentiation of Firms in Technology Space using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The current local batch contains 5,192 observations across 443 entities, and the batch headline estimate is -0.193 for Technology differentiation with p = 0.78. The local headline coefficient on technology differentiation is negative and statistically weak after firm and year effects are imposed. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

**Keywords:** technology differentiation; firm performance; patent portfolios; local calculated replication

## 1 Introduction

This replication begins from a paper-first reading of Position and Differentiation of Firms in Technology Space and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The motivating question is: The source paper asks whether technologically differentiated firms perform better than their peers (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The study is worth revisiting because The replica-

tion matters because it tests whether a central strategy result about technological differentiation survives when the semantic layer is stripped back to a fully executable local panel. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The executable evidence base used in this manuscript contains 5,192 observations across 443 entities and spans 1991 to 2015 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (Arts et al., 2023; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper's contribution is a transparent calculated analogue: it preserves the peer-comparison logic and performance outcome while making the measurement substitution explicit. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (Arts et al., 2023; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper sits in a top-journal conversation about exploration, recombinant search, and boundary-spanning differentiation in technological space. (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Cohen and Levinthal, 1990; Kogut and Zander, 1992, 1996; Teece et al., 1997; Benner and Tushman, 2003). Those studies matter here because the present replication asks whether a performance effect attributed to technological differentiation survives when the semantic portfolio layer is replaced by a processed class-based panel. (Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Henderson and Clark, 1990; Anderson and Tushman, 1986; Hargadon and Sutton, 1997; Tripsas and Gavetti, 2000; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Stuart, 2000). The present manuscript therefore treats the source paper as one contribution inside a broader top-journal stream on exploration, recombination, performance measurement, alliance-based learning, inventive search, technological discontinuity, and patent-based inference. The literature review is intentionally broad because each replication paper should stand on a serious journal-level reference

base rather than on the source article alone. (Rosenkopf and Almeida, 2003; Schilling and Phelps, 2007; Almeida and Kogut, 1999; Fleming et al., 2007; Jaffe, 1986; Jaffe et al., 1993; Trajtenberg, 1990; Hall et al., 2005; Henderson and Cockburn, 1994). Because this paper links technology-space positioning to firm-level performance, the literature review also pulls in classic economic work on patents as indicators, top-journal evidence on the market consequences of knowledge creation, and adjacent strategy studies on exploration-performance tradeoffs in technological search. (Griliches, 1990; Jaffe, 1989). A second paper-specific literature block is added so that the manuscript's topic-matched references are carried into the final bibliography explicitly rather than being silently implied by the broader shared review. (Fleming and Sorenson, 2004; Nerkar, 2003). A second paper-specific literature block is added so that the manuscript's topic-matched references are carried into the final bibliography explicitly rather than being silently implied by the broader shared review. (Lavie et al., 2011).

## 2 Original Study, Theory, and Replication Boundary

The focal source article is (Arts et al., 2023). The original study can be summarized as a mechanism paper with a sharp empirical core. The published paper builds technology-space measures from patent text and studies differentiation among U.S. public firms. The main outcome is firm performance, especially Tobin's Q. Technology differentiation is defined relative to technologically similar peers in the same year. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication boundary is therefore explicit. Di-

rect source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

## 2.1 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The source paper uses semantic patent content. The local reconstruction therefore adopts the following bounded decision: The local rerun uses processed primary-USPC class portfolios. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Tobin's Q is a central outcome. The local reconstruction therefore adopts the following bounded decision: The local panel keeps Tobin's Q because it is fully reconstructible. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Differentiation is relative to peers. The local reconstruction therefore adopts the following bounded decision: Peer comparison is preserved with a class-based similarity measure. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built

from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

| DirectSourceFact | LocalReplicationDecision |
|---|---|
| The source paper uses semantic patent content. | The local rerun uses processed primary-USPC class portfolios. |
| Tobin's Q is a central outcome. | The local panel keeps Tobin's Q because it is fully reconstructible. |
| Differentiation is relative to peers. | Peer comparison is preserved with a class-based similarity measure. |

Direct source facts and explicit local execution decisions carried in the body of the paper.

## 3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are DISCERN-linked firm-patent assignments with unambiguous GVKEYs. PatentView-derived utility-patent and primary-USPC class information. Compustat annual fundamentals used to reconstruct Tobin's Q and controls (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 5,192 observations across 443 entities and covers 1991 to 2015 in the local executable panel. Every numeric claim in the results section is intended to trace back to one of those verified evidence layers rather than to a hand-copied number from the article (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from

the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The local rerun replaces the paper's text-semantic portfolio construction with a class-based five-year rolling USPC portfolio built from processed patent-firm assignments. The local sample frame is Processed public-firm GVKEY-year panel with patent portfolios and financial controls The observed time window in the executable panel is 1991 to 2015 in the local executable panel The outcome is Winsorized Tobin's Q at the firm-year level (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function (Arts et al., 2023; Hall et al., 2005).

### 3.1 Executable File Selection and Data Availability

The present manuscript works from the processed datasets and calculated exports already available in the workspace for Arts, Cassiman, and Hou (2023). That does not mean the paper skips data analysis. It means the data analysis begins from the current executable panel, current coefficient export, and current sample-summary files rather than from a freshly rebuilt raw archive. The substantive obligation is therefore to explain exactly why each verified file is needed and how it enters the paper's argument (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). The analysis panel used in this paper serves a specific empirical role: it is needed to estimate the local model and regenerate descriptive statistics. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The coefficient export used in this paper serves a specific empirical role: it is needed to recover the reported local coefficients and p-values. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The sample summary

used in this paper serves a specific empirical role: it is needed to audit sample construction and raw grouped moments. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The fit summary used in this paper serves a specific empirical role: it is needed to check model diagnostics when a fit file is available. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, these files form the minimum executable evidence chain behind the paper. The theory comes from the source article, the estimable sample comes from the processed analysis panel, the regression narrative comes from the current coefficient export, and the sample audit comes from the local summary file. That is the basis on which the manuscript makes every empirical claim (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

| Stage | NeededFor | Exists | Status |
|---|---|---|---|
| Source article | Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices | Yes | Direct source |
| Analysis panel | Estimate the local model and regenerate descriptive statistics | Yes | Executable local dataset |
| Coefficient export | Recover the reported local coefficients and p-values | Yes | Executable local result file |
| Sample summary | Audit sample construction and raw grouped moments | Yes | Executable local result file |

| Stage | NeededFor | Exists | Status |
|---|---|---|---|
| Fit summary | Check model diagnostics when a fit file is available | Yes | Executable local result file |

Verified evidence layers used to support the paper's calculations.

### 3.2 Sample Reconstruction

The local analysis file contains 5,192 observations drawn from 443 entities. The observed time span runs from 1991 to 2015. Selected sample moments are Winsorized Tobin's Q = 2.177, Technology Differentiation = 0.668 (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

| Statistic | Value |
|---|---|
| Observations | 5,192 |
| Unique Firms | 443 |
| Start Year | 1991 |
| End Year | 2015 |
| Mean Winsorized Tobin's Q | 2.177 |
| Mean Technology Differentiation | 0.668 |
| SD Technology Differentiation | 0.173 |

Sample reconstruction from the current local dataset.

### 3.3 Variable Construction and Derivation

The construct Technology portfolio is operationalized locally as Five-year rolling primary-USPC portfolio and appears in the executable dataset as portfolio_patents; class_count; class_hhi. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Aggregate utility patents by GVKEY and primary USPC class over t-5 to t-1.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Technology differentiation is operationalized locally as One minus peer similarity and appears in the executable dataset as tech_diff. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Compute one minus average similarity to close peers.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Peer benchmark is operationalized locally as Nearest peers in year t and appears in the executable dataset as neighbor_similarity. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use local class-based portfolios to define the similarity set.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of

the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Firm performance is operationalized locally as Winsorized Tobin's Q and appears in the executable dataset as tobin_q_w. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Reconstruct Tobin's Q from processed Compustat variables and winsorize.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Financial controls is operationalized locally as Assets, leverage, cash, R&D, portfolio size and appears in the executable dataset as log_assets; leverage; cash_ratio; rd_intensity; log1p(portfolio_patents). The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Carry forward the financial controls already built in the local script.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

| Construct | Local measure | Derivation |
|---|---|---|
| Technology portfolio | Five-year rolling primary-USPC portfolio | Aggregate utility patents by GVKEY and primary USPC class over t-5 to t-1. |
| Technology differentiation | One minus peer similarity | Compute one minus average similarity to close peers. |
| Peer benchmark | Nearest peers in year t | Use local class-based portfolios to define the similarity set. |
| Firm performance | Winsorized Tobin's Q | Reconstruct Tobin's Q from processed Compustat variables and winsorize. |
| Financial controls | Assets, leverage, cash, R&D, portfolio size | Carry forward the financial controls already built in the local script. |

Variable mapping and derivation logic in the main text.

### 3.4 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Arts, Cassiman, and Hou (2023). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Technology portfolio, the manuscript uses Five-year rolling primary-USPC portfolio under the exact local implementation portfolio_patents; class_count; class_hhi. The derivation step is: Aggregate utility patents by GVKEY and primary USPC class over t-5 to t-1.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Technology differentiation, the manuscript uses One minus peer similarity under the exact local implementation tech_diff. The derivation step is: Compute one minus average similarity to close peers.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Peer benchmark, the manuscript uses Nearest peers in year t under the exact local implementation neighbor_similarity. The derivation step is: Use local class-based portfolios to define the similarity set.. This construct is retained because it occupies the same analytic position in the local design

that the published paper assigns to the original concept. For Firm performance, the manuscript uses Winsorized Tobin's Q under the exact local implementation tobin_q_w. The derivation step is: Reconstruct Tobin's Q from processed Compustat variables and winsorize.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Financial controls, the manuscript uses Assets, leverage, cash, R&D, portfolio size under the exact local implementation log_assets; leverage; cash_ratio; rd_intensity; log1p(portfolio_patents). The derivation step is: Carry forward the financial controls already built in the local script.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

## 4 Descriptive Analysis of the Current Dataset

The current dataset yields a richer descriptive picture than a single mean or coefficient can convey. Winsorized Tobin's Q has mean 2.177, standard deviation 1.508, and an interquartile range from 1.319 to 2.464; Technology differentiation has mean 0.668, standard deviation 0.173, and an interquartile range from 0.558 to 0.800; Neighbor similarity has mean 0.332, standard deviation 0.173, and an interquartile range from 0.200 to 0.442; Portfolio patents has mean 552.7, standard deviation 1,484, and an interquartile range from 29.00 to 450.0 (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The time-series view is equally informative. Observations changes from 233.0 in the first observed period to 204.0 in the last observed period; MeanTechDiff changes from 0.670 in the first observed period to 0.630 in the last observed period; MeanTobinQ changes from 1.722 in the first observed period to 2.465 in the last observed period . These temporal shifts help explain why a paper-comparable replication has to describe the support of the data before turning to regression output (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The regressor-quantile analysis shows that mean outcomes range from 1.957 to 2.857 across the support of the focal regressor, while the mean regressor itself ranges from 0.401 to 0.883. That pattern

is useful because it shows what the formal model is trying to account for in the current dataset (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The correlation structure is reported in a compact form because it helps distinguish whether the main explanatory variables move in lockstep with the controls or whether the regression is drawing on independent variation. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

### 4.1 Data Quality, Coverage, and Support

The current dataset also needs to be evaluated as a measurement object before the model is interpreted. R&D intensity has missing share 9.77%; Winsorized Tobin's Q has missing share 0.00%; Technology differentiation has missing share 0.00% . At the other end of the support, Winsorized Tobin's Q remains close to complete with missing share 0.00%; Technology differentiation remains close to complete with missing share 0.00%; Neighbor similarity remains close to complete with missing share 0.00% . This matters because the credibility of the regression depends on whether the focal constructs survive sample cleaning with enough support to be interpreted substantively (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). Coverage across entities is also visible in the current panel rather than assumed. The five most heavily observed entities contribute between 23 and 23 observations each. The purpose of reporting this concentration pattern is to show whether the estimated mechanism is being learned from a broad panel or from a narrow set of heavily observed entities (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). The support of the focal regressor is broad enough to permit a distributional reading in addition to the formal model. In the lowest observed quantile, the mean regressor is 0.401 and the mean outcome is 2.857 across 1,039 observations. In the highest observed quantile, the corresponding values are 0.883, 1.957, and 1,039 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). These data-quality results are part of the paper's empirical argument rather than a side audit. The replication matters because it tests whether a central strategy result about

technological differentiation survives when the semantic layer is stripped back to a fully executable local panel. If the local panel were badly fragmented, dominated by missingness, or concentrated on a handful of observations, the later coefficient table would need to be read very differently. Bringing this evidence into the body of the article is therefore necessary for an honest replication (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005).

| Variable | Missing | MissingShare |
|---|---|---|
| Winsorized Tobin's Q | 0 | 0.00% |
| Technology differentiation | 0 | 0.00% |
| Neighbor similarity | 0 | 0.00% |
| Portfolio patents | 0 | 0.00% |
| Class count | 0 | 0.00% |
| Class concentration | 0 | 0.00% |
| Log assets | 0 | 0.00% |
| Leverage | 0 | 0.00% |
| Cash ratio | 0 | 0.00% |
| R&D intensity | 507 | 9.77% |

Missingness profile for variables analyzed in the body of the paper.

| Variable | N | Mean | SD | P25 | Median | P75 | Min | Max |
|---|---|---|---|---|---|---|---|---|
| Winsorized Tobin's Q | 5,192 | 2.177 | 1.508 | 1.319 | 1.725 | 2.464 | 0 | 23 |
| Technology differentiation | 5,192 | 0.668 | 0.173 | 0.558 | 0.694 | 0.800 | 0 | 1 |
| Neighbor similarity | 5,192 | 0.332 | 0.173 | 0.200 | 0.306 | 0.442 | 0 | 1 |
| Portfolio patents | 5,192 | 552.7 | 1,484 | 29.00 | 129.0 | 450.0 | 1 | 33760 |
| Class count | 5,192 | 41.74 | 42.92 | 10.00 | 28.00 | 57.00 | 1 | 250 |
| Class concentration | 5,192 | 0.213 | 0.220 | 0.0714 | 0.135 | 0.265 | 0 | 1 |
| Log assets | 5,192 | 8.840 | 1.357 | 7.906 | 8.719 | 9.664 | 4 | 14 |
| Leverage | 5,192 | 0.588 | 0.222 | 0.454 | 0.584 | 0.711 | 0 | 3 |
| Cash ratio | 5,192 | 0.135 | 0.147 | 0.0280 | 0.0802 | 0.191 | 0 | 1 |

| Variable | N | Mean | SD | P25 | Median | P75 | Min | Max |
|---|---|---|---|---|---|---|---|---|
| R&D intensity | 4,685 | 0.0519 | 0.0540 | 0.0146 | 0.0327 | 0.0741 | 0 | 1 |

Main descriptive statistics discussed in the body of the paper.

| Period | Observations | MeanTechDiff | MeanTobinQ | MeanPortfolioPatents |
|---|---|---|---|---|
| 1991 | 233 | 0.670 | 1.722 | 321 |
| 1992 | 237 | 0.665 | 1.742 | 326 |
| 1993 | 238 | 0.676 | 1.791 | 338 |
| 1994 | 239 | 0.678 | 1.735 | 350 |
| 1995 | 236 | 0.683 | 2.017 | 367 |
| 1996 | 236 | 0.659 | 2.138 | 381 |
| 1997 | 233 | 0.649 | 2.475 | 404 |
| 1998 | 221 | 0.650 | 2.653 | 440 |
| 1999 | 211 | 0.651 | 2.939 | 488 |
| 2000 | 221 | 0.680 | 2.979 | 520 |
| 2001 | 238 | 0.683 | 2.473 | 544 |
| 2002 | 237 | 0.689 | 2.000 | 527 |
| 2005 | 227 | 0.696 | 2.209 | 268 |
| 2006 | 220 | 0.688 | 2.294 | 280 |
| 2007 | 210 | 0.689 | 2.301 | 374 |
| 2008 | 214 | 0.686 | 1.709 | 517 |
| 2009 | 225 | 0.686 | 1.885 | 641 |
| 2010 | 227 | 0.671 | 2.019 | 737 |
| 2011 | 225 | 0.658 | 1.932 | 811 |
| 2012 | 220 | 0.652 | 1.984 | 891 |
| 2013 | 223 | 0.635 | 2.269 | 974 |
| 2014 | 217 | 0.639 | 2.517 | 1,106 |
| 2015 | 204 | 0.630 | 2.465 | 1,249 |

Observed time patterns in the current dataset.

| Quantile | N | MeanRegressor | MeanOutcome |
|---|---|---|---|
| Q1 | 1,039 | 0.401 | 2.857 |
| Q2 | 1,038 | 0.586 | 2.096 |
| Q3 | 1,038 | 0.693 | 1.997 |
| Q4 | 1,038 | 0.778 | 1.978 |
| Q5 | 1,039 | 0.883 | 1.957 |

Grouped summaries used to describe raw patterns in the current dataset.

| Variable | Winsorized Tobin's Q | Technology differentiation | Neighbor similarity | Portfolio patents | Class count | Class concentration | Log assets | Leverage | Cash ratio | R&D intensity |
|---|---|---|---|---|---|---|---|---|---|---|
| Winsorized Tobin's Q | 1.000 | -0.214 | 0.214 | 0 | 0 | 0.213 | -0.097 | -0.245 | 0.357 | 0.340 |
| Technology differentiation | -0.214 | 1.000 | -1.000 | 0 | 0 | 0.106 | -0.224 | 0.125 | -0.298 | -0.396 |
| Neighbor similarity | 0.214 | -1.000 | 1.000 | 0 | 0 | -0.106 | 0.224 | -0.125 | 0.298 | 0.396 |
| Portfolio patents | -0.029 | -0.221 | 0.221 | 1 | 1 | -0.200 | 0.414 | 0.037 | 0.044 | 0.053 |
| Class count | -0.144 | -0.153 | 0.153 | 1 | 1 | -0.503 | 0.561 | 0.151 | -0.124 | -0.051 |
| Class concentration | 0.213 | 0.106 | -0.106 | 0 | -1 | 1.000 | -0.249 | -0.110 | 0.130 | 0.112 |

Compact correlation structure for the main variables.

# 5 Empirical Strategy and Identification

The estimand targeted in this manuscript is A firm and year fixed-effects regression of winsorized Tobin's Q on local technology differentiation and financial controls. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into

an experimental design, but it does make the inferential claim narrower and more honest (Arts et al., 2023; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The first set of replication decisions concerns scope. The unit of analysis is Firm-year, the sample frame is Processed public-firm GVKEY-year panel with patent portfolios and financial controls, and the empirical window is 1991 to 2015 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The second set of decisions concerns measurement and specification. The outcome is Winsorized Tobin's Q at the firm-year level, and the main explanatory design is Firm and year fixed-effects regression of Tobin's Q on local technology differentiation and controls. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Financial controls plus firm and year fixed effects.. The crucial exclusion or substitution rule is: Original semantic embedding inputs are unavailable locally, so the executable rerun substitutes class-based portfolio similarity and keeps only observations supported by the processed panel.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. When the local data architecture differs, the paper marks the result as a calculated analogue rather than claiming byte-identical duplication (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

| DecisionPoint | LocalSpecification |
|---|---|
| Research question | The source paper asks whether technologically differentiated firms perform better than their peers. |
| Unit of analysis | Firm-year |
| Sample frame | Processed public-firm GVKEY-year panel with patent portfolios and financial controls. |
| Time window | 1991 to 2015 in the local executable panel. |
| Outcome definition | Winsorized Tobin's Q at the firm-year level. |
| Main explanatory design | Firm and year fixed-effects regression of Tobin's Q on local technology differentiation and controls. |
| Controls and fixed effects | Financial controls plus firm and year fixed effects. |
| Exclusion or substitution logic | Original semantic embedding inputs are unavailable locally, so the executable rerun substitutes class-based portfolio similarity and keeps only observations supported by the processed panel. |

Replication decision register retained in the main text.

## 5.1 Why These Design Decisions Are Necessary

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, mea-

surement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Firm-year inside the sample frame Processed public-firm GVKEY-year panel with patent portfolios and financial controls over 1991 to 2015 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Winsorized Tobin's Q at the firm-year level and the main explanatory design is Firm and year fixed-effects regression of Tobin's Q on local technology differentiation and controls. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Financial controls plus firm and year fixed effects. The main exclusion or substitution rule is Original semantic embedding inputs are unavailable locally, so the executable rerun substitutes class-based portfolio similarity and keeps only observations supported by the processed panel. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

$$Q_{it}^{(w)} = \beta_1 \text{TechDiff}_{it} + \beta_2 X_{it} + \alpha_i + \gamma_t + \varepsilon_{it}. \tag{1}$$

# 6 Results

## 6.1 Main Estimates

The central local estimate is -0.193 on Technology differentiation (p = 0.78). The local headline coefficient on technology differentiation is negative and statistically weak after firm and year effects are imposed. Log assets is estimated at -0.516 with p = 0.002. Log(1 + portfolio patents) is estimated at -0.266 with p = 0.002. R&D intensity is estimated at 4.290 with p = 0.009 (Arts et al., 2023; Hall et al., 2005). The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

| Model | Term | Estimate | Std. error | P-value |
|---|---|---|---|---|
| Arts Fe | Technology differentiation | -0.1935 | 0.6974 | 0.784 |
| Arts Fe | Log assets | -0.5161 | 0.1442 | 0.0017 |

| Model | Term | Estimate | Std. error | P-value |
|---|---|---|---|---|
| Arts Fe | Leverage | -0.6522 | 0.2840 | 0.0315 |
| Arts Fe | Cash ratio | 1.565 | 0.6384 | 0.0226 |
| Arts Fe | R&D intensity | 4.290 | 1.505 | 0.0093 |
| Arts Fe | Log(1 + portfolio patents) | -0.2660 | 0.0764 | 0.0021 |
| Arts Fe | Class count | 0.0043 | 0.0024 | 0.0851 |
| Arts Fe | Class concentration | 0.4918 | 0.2947 | 0.109 |

Key coefficient table used in the main text.

## 6.2 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Log assets is estimated at -0.516 with a p-value of 0.002; Log(1 + portfolio patents) is estimated at -0.266 with a p-value of 0.002; R&D intensity is estimated at 4.290 with a p-value of 0.009; Cash ratio is estimated at 1.565 with a p-value of 0.023 (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Within the focal specification, the sharpest local signal comes from Log assets, which is estimated at -0.516 with $p = 0.002$. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The scale of the underlying variables also matters for interpretation. The outcome Winsorized Tobin's Q has mean 2.177 and standard deviation 1.508 in the local panel, while the main regressor Technology differentiation has mean 0.668 and standard deviation 0.173 (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 1.957 to 2.857 while the mean regressor ranges from 0.401 to 0.883. This does not substitute for the model, but it shows how much raw variation the specification must explain (Arts et al., 2023; Fleming, 2001; Ahuja

and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local headline coefficient on technology differentiation is negative and statistically weak after firm and year effects are imposed. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

### 6.3 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within arts_fe, the leading coefficient family reads as follows: Log assets = -0.516 (p = 0.002); Log(1 + portfolio patents) = -0.266 (p = 0.002); R&D intensity = 4.290 (p = 0.009); Cash ratio = 1.565 (p = 0.023); Leverage = -0.652 (p = 0.032). The largest positive estimate in this specification is R&D intensity at 4.290, while the largest negative estimate is Leverage at -0.652. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This fuller coefficient reading is especially important for Arts, Cassiman, and Hou (2023) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text (Arts et al., 2023; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

## 6.4 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Arts Fe, the most informative focal term is Log assets with estimate -0.516 and p = 0.002 (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Arts Fe specification contributes its own reading to the replication. Log assets is estimated at -0.516 with p = 0.002 and is classified as a strong signal; Log(1 + portfolio patents) is estimated at -0.266 with p = 0.002 and is classified as a strong signal; R&D intensity is estimated at 4.290 with p = 0.009 and is classified as a strong signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This model comparison matters for Arts, Cassiman, and Hou (2023) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

| Term | Model | Estimate | PValue | Signal | Reading |
|---|---|---|---|---|---|
| Technology differentiation | Arts Fe | -0.1935 | 0.78 | Weak | Negative estimate with p = 0.78. |
| Log assets | Arts Fe | -0.5161 | 0.002 | Strong | Negative estimate with p = 0.002. |
| Leverage | Arts Fe | -0.6522 | 0.032 | Moderate | Negative estimate with p = 0.032. |
| Cash ratio | Arts Fe | 1.565 | 0.023 | Moderate | Positive estimate with p = 0.023. |

| Term | Model | Estimate | PValue | Signal | Reading |
|---|---|---|---|---|---|
| R&D intensity | Arts Fe | 4.290 | 0.009 | Strong | Positive estimate with p = 0.009. |
| Log(1 + portfolio patents) | Arts Fe | -0.2660 | 0.002 | Strong | Negative estimate with p = 0.002. |
| Class count | Arts Fe | 0.0043 | 0.085 | Suggestive | Positive estimate with p = 0.085. |
| Class concentration | Arts Fe | 0.4918 | 0.11 | Weak | Positive estimate with p = 0.11. |

Model-by-model interpretation table drawn from the current coefficient exports.

## 6.5 Diagnostic Evidence

Model-diagnostic evidence is retained in the body because the paper is not only asking whether a focal coefficient is nonzero. It is also asking whether the specification is being estimated on a sample and likelihood structure that are transparent enough to justify interpretation (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). For arts_fe, the current fit export reports N = 4,669, log-likelihood = -6,269, AIC = 13,390, BIC = 16,137, within R-squared = 0.144 (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005). These diagnostics do not, by themselves, validate the source paper's theory. They do, however, show that the local data analysis behind Arts, Cassiman, and Hou (2023) is being carried out on a clearly described estimating sample with observable fit statistics (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005).

| n | ll | aic | bic | wr2 | model |
|---|---|---|---|---|---|
| 4,669 | -6,269 | 13,390 | 16,137 | 0.144 | arts_fe |

Model fit statistics discussed in the main results section.

| Model | Term | Estimate | Std. error | Statistic | P-value |
|---|---|---|---|---|---|
| Arts Fe | Technology differentiation | -0.1935 | 0.6974 | -0.277 | 0.784 |
| Arts Fe | Log assets | -0.5161 | 0.1442 | -3.579 | 0.0017 |
| Arts Fe | Leverage | -0.6522 | 0.2840 | -2.297 | 0.0315 |
| Arts Fe | Cash ratio | 1.565 | 0.6384 | 2.452 | 0.0226 |
| Arts Fe | R&D intensity | 4.290 | 1.505 | 2.850 | 0.0093 |
| Arts Fe | Log(1 + portfolio patents) | -0.2660 | 0.0764 | -3.480 | 0.0021 |
| Arts Fe | Class count | 0.0043 | 0.0024 | 1.803 | 0.0851 |
| Arts Fe | Class concentration | 0.4918 | 0.2947 | 1.669 | 0.109 |

Detailed coefficient table from the current local estimation output.

# 7 Comparison to the Published Study

## 7.1 Original Results Versus the Current Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central empirical claim, the main measurement layer, and the effective sample definition are set alongside the current local dataset result so the reader can see where agreement is substantive and where it is conditional on different data architecture. (Arts et al., 2023). For Arts, Cassiman, and Hou (2023), the local headline result is Technology differentiation = -0.193 with p = 0.78. The value of presenting that estimate directly against the original paper's reported finding is that it shows whether the mechanism survives a different but transparent processed dataset rather than asking the reader to infer comparability from prose alone.

| Aspect | OriginalPaper | LocalDataset | Reading |
|---|---|---|---|
| Main empirical result | The published paper reports that greater technological differentiation is associated with stronger firm performance when firms are compared to technologically similar peers. | Technology differentiation = -0.193 with p = 0.78. | The source paper reports a positive performance relationship, whereas the local executable analogue produces a negative and imprecise coefficient. That gap is useful evidence about how much the published result may depend on the original semantic measurement layer and sample construction. |
| Measurement and data construction | The published measure is built from patent text and semantic technology-space positioning. | The local rerun replaces the paper's text-semantic portfolio construction with a class-based five-year rolling USPC portfolio built from processed patent-firm assignments. | This row isolates how the executable local dataset differs from the original measurement architecture while preserving the same substantive target. |
| Sample and unit of analysis | The published design studies U.S. public firms in a firm-year panel with peer-relative positioning. | Processed public-firm GVKEY-year panel with patent portfolios and financial controls. Local sample = 5,192 observations across 443 entities. | This row clarifies whether agreement or disagreement with the original paper could reflect sample translation rather than theory alone. |

Comparison of the original paper's results and design to the current local dataset analysis.

## 7.2 Interpretive Implications of the Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published measure is built from patent text and semantic technology-space positioning By contrast, the local paper relies on the following executable design: The local rerun replaces the paper's text-semantic portfolio construction with a class-based five-year rolling USPC portfolio built from processed patent-firm assignments These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The same caution applies to the sample itself. The published design is described as follows: The published design studies U.S. public firms in a firm-year panel with peer-relative positioning The local sample instead works from Processed public-firm GVKEY-year panel with patent portfolios and financial controls and yields 5,192 observations across 443 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable ways. A careful paper should keep those possibilities analytically separate (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. The source paper reports a positive performance relationship, whereas the local executable analogue produces a negative and imprecise coefficient. That gap is useful evidence about how much the published result may depend on the original semantic measurement layer and sample construction. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert,

2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The comparison to the published paper is intentionally narrow. The source paper reports a positive performance relationship, whereas the local executable analogue produces a negative and imprecise coefficient. That gap is useful evidence about how much the published result may depend on the original semantic measurement layer and sample construction. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

# 8 Discussion

The source paper reports a positive performance relationship, whereas the local executable analogue produces a negative and imprecise coefficient. That gap is useful evidence about how much the published result may depend on the original semantic measurement layer and sample construction (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly (Arts et al., 2023; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The first limitation concerns measurement. The exact text-embedding pipeline from the source paper is not stored locally (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The second limitation concerns sample

translation. The local differentiation measure is therefore class-based rather than text-semantic (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The third limitation concerns interpretation. Comparisons to the published estimates should be read as inferential rather than byte-identical (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005).

## 9 Conclusion

The paper's contribution is a transparent calculated analogue: it preserves the peer-comparison logic and performance outcome while making the measurement substitution explicit (Arts et al., 2023; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (Arts et al., 2023; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

# A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

| Statistic | Value | RawStatistic |
|---|---:|---:|
| Observations | 5,192 | rows |
| Unique Firms | 443 | unique_firms |
| Start Year | 1991 | year_min |
| End Year | 2015 | year_max |
| Mean Winsorized Tobin's Q | 2.177 | mean_tobin_q_w |
| Mean Technology Differentiation | 0.668 | mean_tech_diff |
| SD Technology Differentiation | 0.173 | sd_tech_diff |

Detailed sample reconstruction table.

# B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

| Variable | N | Mean | SD | P25 | Median | P75 | Min | Max |
|---|---|---|---|---|---|---|---|---|
| Winsorized Tobin's Q | 5,192 | 2.177 | 1.508 | 1.319 | 1.725 | 2.464 | 0 | 23 |
| Technology differentiation | 5,192 | 0.668 | 0.173 | 0.558 | 0.694 | 0.800 | 0 | 1 |
| Neighbor similarity | 5,192 | 0.332 | 0.173 | 0.200 | 0.306 | 0.442 | 0 | 1 |
| Portfolio patents | 5,192 | 552.7 | 1,484 | 29.00 | 129.0 | 450.0 | 1 | 33760 |
| Class count | 5,192 | 41.74 | 42.92 | 10.00 | 28.00 | 57.00 | 1 | 250 |
| Class concentration | 5,192 | 0.213 | 0.220 | 0.0714 | 0.135 | 0.265 | 0 | 1 |
| Log assets | 5,192 | 8.840 | 1.357 | 7.906 | 8.719 | 9.664 | 4 | 14 |
| Leverage | 5,192 | 0.588 | 0.222 | 0.454 | 0.584 | 0.711 | 0 | 3 |
| Cash ratio | 5,192 | 0.135 | 0.147 | 0.0280 | 0.0802 | 0.191 | 0 | 1 |
| R&D intensity | 4,685 | 0.0519 | 0.0540 | 0.0146 | 0.0327 | 0.0741 | 0 | 1 |

Descriptive statistics appendix.

# C Appendix C: Correlation Matrix

| Variable | Winsorized Tobin's Q | Technology differentiation | Neighbor similarity | Portfolio patents | Class count | Class concentration | Log assets | Leverage | Cash ratio | R&D intensity |
|---|---|---|---|---|---|---|---|---|---|---|
| Winsorized Tobin's Q | 1.000 | -0.214 | 0.214 | 0 | 0 | 0.213 | -0.097 | -0.245 | 0.357 | 0.340 |
| Technology differentiation | -0.214 | 1.000 | -1.000 | 0 | 0 | 0.106 | -0.224 | 0.125 | -0.298 | -0.396 |
| Neighbor similarity | 0.214 | -1.000 | 1.000 | 0 | 0 | -0.106 | 0.224 | -0.125 | 0.298 | 0.396 |
| Portfolio patents | -0.029 | -0.221 | 0.221 | 1 | 1 | -0.200 | 0.414 | 0.037 | 0.044 | 0.053 |
| Class count | -0.144 | -0.153 | 0.153 | 1 | 1 | -0.503 | 0.561 | 0.151 | -0.124 | -0.051 |
| Class concentration | 0.213 | 0.106 | -0.106 | 0 | -1 | 1.000 | -0.249 | -0.110 | 0.130 | 0.112 |
| Log assets | -0.097 | -0.224 | 0.224 | 0 | 1 | -0.249 | 1.000 | 0.140 | -0.133 | -0.205 |
| Leverage | -0.245 | 0.125 | -0.125 | 0 | 0 | -0.110 | 0.140 | 1.000 | -0.364 | -0.315 |
| Cash ratio | 0.357 | -0.298 | 0.298 | 0 | 0 | 0.130 | -0.133 | -0.364 | 1.000 | 0.513 |
| R&D intensity | 0.340 | -0.396 | 0.396 | 0 | 0 | 0.112 | -0.205 | -0.315 | 0.513 | 1.000 |

Correlation matrix for the main numeric variables.

# D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

| Period | Observations | MeanTechDiff | MeanTobinQ | MeanPortfolioPatents |
|---|---|---|---|---|
| 1991 | 233 | 0.670 | 1.722 | 321 |
| 1992 | 237 | 0.665 | 1.742 | 326 |
| 1993 | 238 | 0.676 | 1.791 | 338 |
| 1994 | 239 | 0.678 | 1.735 | 350 |
| 1995 | 236 | 0.683 | 2.017 | 367 |
| 1996 | 236 | 0.659 | 2.138 | 381 |
| 1997 | 233 | 0.649 | 2.475 | 404 |
| 1998 | 221 | 0.650 | 2.653 | 440 |
| 1999 | 211 | 0.651 | 2.939 | 488 |
| 2000 | 221 | 0.680 | 2.979 | 520 |
| 2001 | 238 | 0.683 | 2.473 | 544 |
| 2002 | 237 | 0.689 | 2.000 | 527 |
| 2005 | 227 | 0.696 | 2.209 | 268 |
| 2006 | 220 | 0.688 | 2.294 | 280 |
| 2007 | 210 | 0.689 | 2.301 | 374 |
| 2008 | 214 | 0.686 | 1.709 | 517 |
| 2009 | 225 | 0.686 | 1.885 | 641 |
| 2010 | 227 | 0.671 | 2.019 | 737 |
| 2011 | 225 | 0.658 | 1.932 | 811 |
| 2012 | 220 | 0.652 | 1.984 | 891 |
| 2013 | 223 | 0.635 | 2.269 | 974 |
| 2014 | 217 | 0.639 | 2.517 | 1,106 |
| 2015 | 204 | 0.630 | 2.465 | 1,249 |

Year-by-year sample audit.

# E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

| Quantile | N | MeanRegressor | MeanOutcome |
|---|---|---|---|
| Q1 | 1,039 | 0.401 | 2.857 |
| Q2 | 1,038 | 0.586 | 2.096 |
| Q3 | 1,038 | 0.693 | 1.997 |
| Q4 | 1,038 | 0.778 | 1.978 |
| Q5 | 1,039 | 0.883 | 1.957 |

Quantile summary for the main regressor.

# F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

| term | estimate | std_error | statistic | p_value | model |
|---|---|---|---|---|---|
| tech_diff | -0.1935 | 0.6974 | -0.277 | 0.78 | arts_fe |
| log_assets | -0.5161 | 0.1442 | -3.579 | 0.002 | arts_fe |
| leverage | -0.6522 | 0.2840 | -2.297 | 0.032 | arts_fe |
| cash_ratio | 1.565 | 0.6384 | 2.452 | 0.023 | arts_fe |
| rd_intensity | 4.290 | 1.505 | 2.850 | 0.009 | arts_fe |
| log1p(portfolio_patents) | -0.2660 | 0.0764 | -3.480 | 0.002 | arts_fe |
| class_count | 0.0043 | 0.0024 | 1.803 | 0.085 | arts_fe |
| class_hhi | 0.4918 | 0.2947 | 1.669 | 0.11 | arts_fe |

Full coefficient dump exported from the local estimation script.

# G Appendix G: Coverage Tables

| Variable | Missing | MissingShare |
|---|---|---|
| Winsorized Tobin's Q | 0 | 0.00% |
| Technology differentiation | 0 | 0.00% |
| Neighbor similarity | 0 | 0.00% |
| Portfolio patents | 0 | 0.00% |
| Class count | 0 | 0.00% |
| Class concentration | 0 | 0.00% |
| Log assets | 0 | 0.00% |
| Leverage | 0 | 0.00% |
| Cash ratio | 0 | 0.00% |
| R&D intensity | 507 | 9.77% |

Missingness audit for the variables summarized in the manuscript.

| Entity | Observations | FirstPeriod | LastPeriod |
|---|---|---|---|
| 10016 | 23 | 1991 | 2015 |
| 10405 | 23 | 1991 | 2015 |
| 10499 | 23 | 1991 | 2015 |
| 10519 | 23 | 1991 | 2015 |
| 1078 | 23 | 1991 | 2015 |
| 10983 | 23 | 1991 | 2015 |
| 10984 | 23 | 1991 | 2015 |
| 11456 | 23 | 1991 | 2015 |
| 11465 | 23 | 1991 | 2015 |
| 1161 | 23 | 1991 | 2015 |
| 11636 | 23 | 1991 | 2015 |
| 1209 | 23 | 1991 | 2015 |
| 1300 | 23 | 1991 | 2015 |

| Entity | Observations | FirstPeriod | LastPeriod |
|---|---|---|---|
| 1356 | 23 | 1991 | 2015 |
| 1690 | 23 | 1991 | 2015 |
| 1722 | 23 | 1991 | 2015 |
| 1913 | 23 | 1991 | 2015 |
| 1988 | 23 | 1991 | 2015 |
| 2044 | 23 | 1991 | 2015 |
| 2086 | 23 | 1991 | 2015 |

Top entities by local observation count.

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